PROJECT REPORT

Car Rental Request Automation

1. INTRODUCTION

1.1 Project Overview

Car Rental Request Automation is a ServiceNow-based project designed to automate the process of submitting, processing, approving, and tracking car rental requests. The project uses Service Catalog and Workflow to replace manual request handling with a structured and automated process.

1.2 Purpose

The main purpose is to provide employees with a simple way to request a rental car through the ServiceNow Service Catalog. The workflow automatically routes the request, manages approval, creates the required tasks, sends notifications, and updates the request status.

2. IDEATION PHASE

2.1 Problem Statement

Manual car rental requests can involve emails, phone calls, repeated data entry, delayed approvals, and difficulty tracking request status. There is a need for a centralized system that standardizes the request process and reduces manual work.

2.2 Empathy Map Canvas

- User needs a simple form to submit car rental requirements.
- User wants to know whether the request is approved and its current status.
- Approver needs complete request information to make a decision.
- Support or rental team needs clear tasks and request details.

2.3 Brainstorming

- Create a Car Rental Request catalog item.
- Capture employee, rental date, location, vehicle type, and purpose.
- Automate approval through ServiceNow Workflow.
- Create fulfillment tasks after approval.
- Send notifications for submission, approval, rejection, and completion.
- Allow request tracking from ServiceNow.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Request Creation → Request Submission → Manager/Authorized Approval → Fulfillment Task → Rental Processing → Request Completion. If the request is rejected, the workflow updates the request accordingly and notifies the requester.

3.2 Solution Requirement

- ServiceNow instance with Service Catalog and Workflow capabilities.
- Catalog item for submitting a car rental request.
- Variables for required rental information.
- Workflow for approval and fulfillment.
- Notifications for important request status changes.
- Request and task records for tracking and audit purposes.

3.3 Data Flow Diagram

Requester → Car Rental Catalog Item → ServiceNow Request/RITM → Workflow → Approval → Fulfillment Task → Rental Team → Completion/Notification.

3.4 Technology Stack

- Platform: ServiceNow
- Service Catalog: Used to create and submit the car rental request.
- Workflow: Used to automate approval, task creation, notifications, and status movement.
- ServiceNow Tables: Request/Requested Item and task-related records.
- ServiceNow Scripting/Configuration: Used where required for validation and automation.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution centralizes the complete car rental request process in ServiceNow. It reduces dependency on manual communication, provides consistent request information, improves visibility, and makes the approval and fulfillment process easier to track.

4.2 Proposed Solution

A Car Rental Request catalog item is created in ServiceNow. The requester enters the required details and submits the request. A workflow receives the request, sends it for approval, and based on the decision either stops the request with a rejection notification or continues to fulfillment. After approval, the required task is assigned to the responsible team. Once the task is completed, the request is closed and the requester is notified.

4.3 Solution Architecture

- Requester Layer: Employee accesses the Service Catalog and submits the car rental request.
- Catalog Layer: Car Rental Request item collects and validates required information.
- Workflow Layer: Controls approval, decision, task creation, notifications, and completion.
- Approval Layer: Authorized approver reviews and approves/rejects the request.
- Fulfillment Layer: Rental/support team receives and completes the generated task.
- Tracking Layer: ServiceNow request records maintain status and history.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Phase 1: Identify the manual car rental request process and required information.

- Phase 2: Design the Service Catalog item and request variables.
- Phase 3: Configure the workflow and approval process.
- Phase 4: Configure task assignment and notifications.
- Phase 5: Test approval, rejection, fulfillment, and completion scenarios.
- Phase 6: Review the complete automated request flow and document the project.

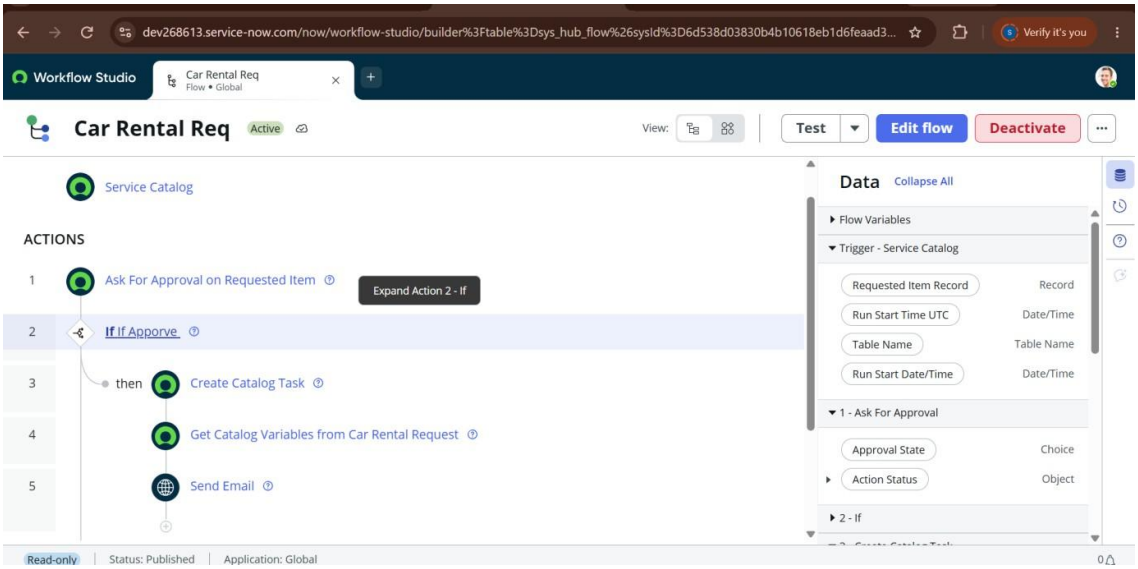
6. FUNCTIONAL AND PERFORMANCE TESTING

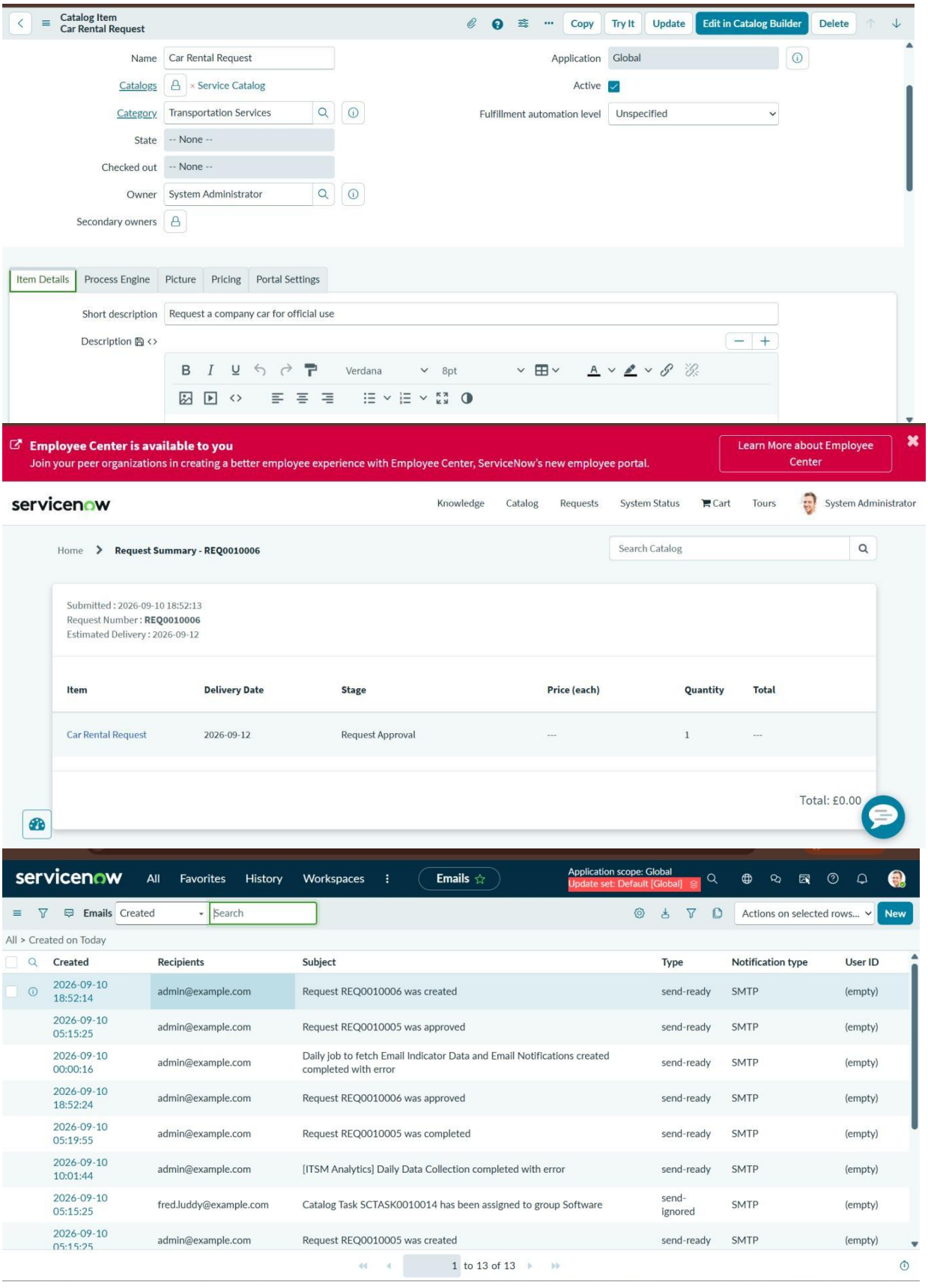
6.1 Performance Testing

Functional testing focuses on whether the catalog item accepts valid request information, required fields are enforced, the workflow starts correctly, approval decisions are handled correctly, fulfillment tasks are created after approval, and notifications/status updates occur as expected. Performance is evaluated from the perspective of smooth form submission and timely workflow processing within the ServiceNow environment.

- Test valid car rental request submission.
- Test missing mandatory information.
- Test approval path.
- Test rejection path.
- Test fulfillment task creation.
- Test request completion and status update.
- Test requester notifications.

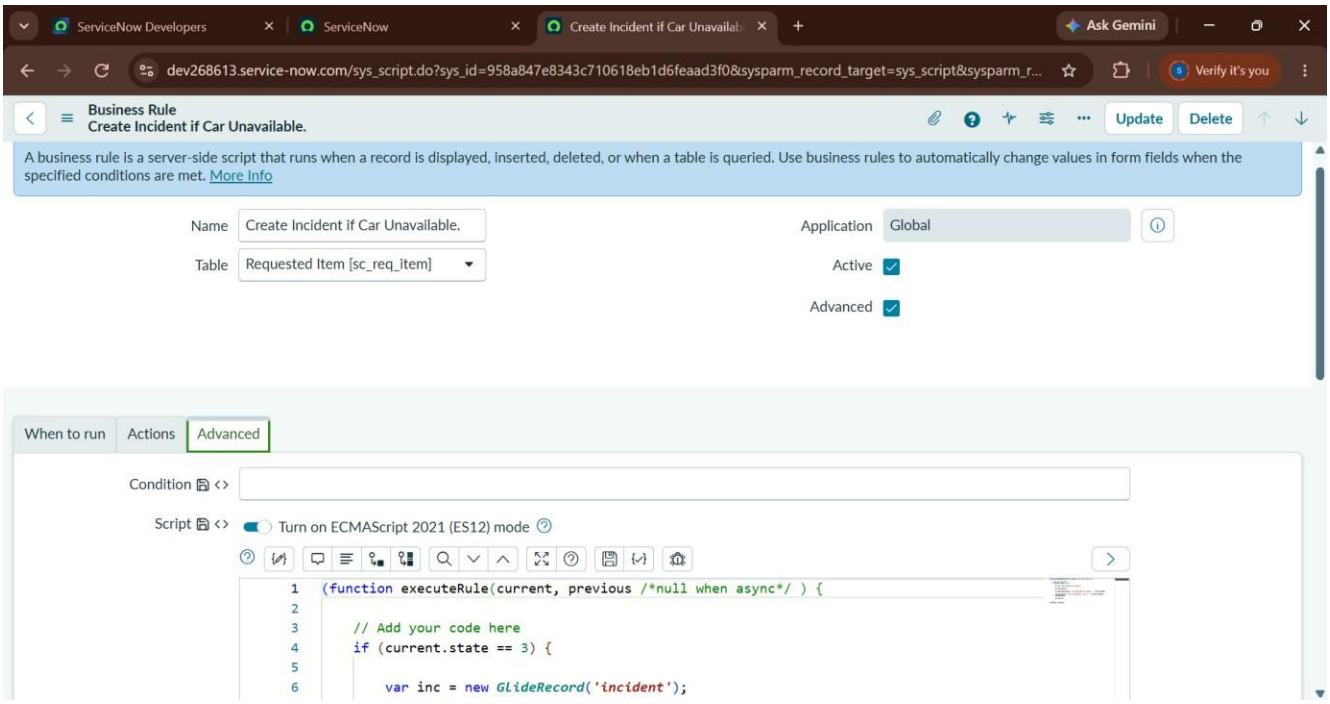
7. RESULTS




Catalog Item
Car Rental Request
Copy
Try It
Update
Edit in Catalog Builder
Delete
Name
Car Rental Request
Application
Global
Catalogs
Service Catalog
Active
Category
Transportation Services
Fulfillment automation level
Unspecified
State
-- None --
Checked out
-- None --
Owner
System Administrator
Secondary owners
Item Details
Process Engine
Picture
Pricing
Portal Settings
Short description
Request a company car for official use
Description
Verdana
8pt
Employee Center is available to you
Join your peer organizations in creating a better employee experience with Employee Center, ServiceNow's new employee portal.
Learn More about Employee Center
servicenow
Knowledge
Catalog
Requests
System Status
Cart
Tours
System Administrator
Home
Request Summary - REQ0010006
Search Catalog
Submitted : 2026-09-10 18:52:13
Request Number : REQ0010006
Estimated Delivery : 2026-09-12
Item
Delivery Date
Stage
Price (each)
Quantity
Total
Car Rental Request
2026-09-12
Request Approval

1

Total: £0.00
servicenow
All
Favorites
History
Workspaces
Emails
Application scope: Global
Update set: Default [Global]
Emails
Created
Search
Actions on selected rows...
New
All > Created on Today
Created
Recipients
Subject
Type
Notification type
User ID
2026-09-10 18:52:14
admin@example.com
Request REQ0010006 was created
send-ready
SMTP
(empty)
2026-09-10 05:15:25
admin@example.com
Request REQ0010005 was approved
send-ready
SMTP
(empty)
2026-09-10 00:00:16
admin@example.com
Daily job to fetch Email Indicator Data and Email Notifications created completed with error
send-ready
SMTP
(empty)
2026-09-10 18:52:24
admin@example.com
Request REQ0010006 was approved
send-ready
SMTP
(empty)
2026-09-10 05:19:55
admin@example.com
Request REQ0010005 was completed
send-ready
SMTP
(empty)
2026-09-10 10:01:44
admin@example.com
[ITSM Analytics] Daily Data Collection completed with error
send-ready
SMTP
(empty)
2026-09-10 05:15:25
fred.luddy@example.com
Catalog Task SCTASK0010014 has been assigned to group Software
send-ignored
SMTP
(empty)
2026-09-10 05:15:25
admin@example.com
Request REQ0010005 was created
send-ready
SMTP
(empty)
1 to 13 of 13



8. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces manual work and repeated communication.
- Provides a centralized request and tracking process.
- Automates approval and fulfillment activities.
- Improves visibility of request status.
- Creates a consistent and auditable process.

Disadvantages

- Requires initial ServiceNow configuration and workflow setup.
- Changes to business rules may require workflow updates.
- The solution depends on the availability and configuration of the ServiceNow platform.

9. CONCLUSION

The Car Rental Request Automation project demonstrates how ServiceNow Service Catalog and Workflow can be used to automate a common business request process. The solution provides a structured flow from request submission to approval and fulfillment, reducing manual effort and improving request visibility and control.

10. FUTURE SCOPE

- Integrate with external car rental or fleet management systems.
- Add automatic vehicle availability checking.
- Add cost estimation and budget validation.
- Support multiple vehicle categories and rental policies.

- Add dashboards and reports for request volume, approval time, and fulfillment performance.
- Add automated reminders for pending approvals and overdue fulfillment tasks.

11. APPENDIX

https://github.com/NVkishore22/Car-Rental-Request-Automation-ServiceNow

FUNCTIONAL SPECIFICATION DOCUMENT (FSD)

Car Rental Request Automation

1. Introduction

Project Title: Car Rental Request Automation

Platform: ServiceNow

Main ServiceNow Components: Service Catalog and Workflow

Document Purpose: This Functional Specification Document describes the functional requirements, process flow, roles, data captured, approvals, fulfillment activities, notifications, and expected behavior of the Car Rental Request Automation solution.

2. Project Overview

2.1 Purpose

The purpose of this project is to automate the car rental request process in ServiceNow. Employees can submit a request through a Service Catalog item, provide the required rental details, and send the request through an automated approval and fulfillment workflow.

2.2 Objectives

- Provide a centralized method for submitting car rental requests.
- Capture all required rental information in a standard format.
- Automatically route requests for approval.
- Create fulfillment tasks after approval.
- Notify users about important request status changes.
- Provide visibility and tracking throughout the request lifecycle.

2.3 Key Features

- Car Rental Request catalog item.
- Mandatory field validation.
- Automated approval workflow.
- Approval and rejection handling.
- Automatic fulfillment task creation.
- Task assignment to the responsible team.
- Requester notifications.
- Request status tracking and audit history.

3. Functional Requirements

FR-01: Request Submission

The system shall allow an authorized requester to open the Car Rental Request catalog item and submit a new rental request.

FR-02: Request Information

The catalog item shall collect required information such as requester, rental date, pickup location, return date, return location, vehicle type, purpose, and other business-required details.

FR-03: Validation

The system shall validate mandatory fields before allowing the request to be submitted.

FR-04: Request Creation

On submission, ServiceNow shall create the required request/requested-item records and initiate the configured workflow.

FR-05: Approval

The workflow shall route the request to the designated approver for review.

FR-06: Approval Decision

If approved, the workflow shall continue to fulfillment. If rejected, the request shall follow the rejection path and the requester shall be notified.

FR-07: Fulfillment

For an approved request, the workflow shall create a fulfillment task and assign it to the responsible rental/support team.

FR-08: Completion

After fulfillment is completed, the request shall be updated to the appropriate completed/closed state.

FR-09: Notifications

The system shall send notifications for relevant events such as submission, approval, rejection, and completion.

FR-10: Tracking

The requester and authorized users shall be able to track the request status and view relevant request information.

4. System Architecture

4.1 ServiceNow Architecture

The solution is implemented inside the ServiceNow platform. The Service Catalog provides the user-facing request form. The submitted catalog request generates ServiceNow request records. Workflow controls the business process, including approval, decision handling, task creation, notifications, and completion.

4.2 Main Components

- Service Catalog Item – Car Rental Request.
- Catalog Variables – Capture rental and requester information.
- Workflow – Automates the request lifecycle.
- Approval Activity – Routes the request to an authorized approver.
- Task Activity – Creates work for the responsible fulfillment team.
- Notification – Communicates important status changes.
- Request/RITM and Task Records – Maintain tracking and history.

5. Service Catalog Specification

5.1 Catalog Item

Name: Car Rental Request

Purpose: Provides employees with a standard form to request a rental vehicle.

5.2 Suggested Catalog Variables

- Requested For – Employee for whom the rental is required.
- Rental Start Date – Date on which the rental begins.
- Rental End Date – Date on which the rental ends.
- Pickup Location – Location where the vehicle is required.
- Return Location – Location where the vehicle will be returned.
- Vehicle Type – Required vehicle category.
- Rental Purpose – Business purpose or reason for the rental.
- Additional Requirements – Optional comments or special requirements.

5.3 Validation Rules

- Mandatory fields must be completed before submission.
- Rental end date should not be earlier than the rental start date.
- Date and location information should follow the configured business rules.
- Only permitted vehicle types should be selectable.

6. Workflow Specification

6.1 Workflow Flow

Request Submission → Workflow Trigger → Approval → Decision → Fulfillment Task → Task Completion → Request Completion → Notification

6.2 Approval Path

After the request is submitted, the workflow sends an approval request to the configured approver. The approver reviews the rental details and either approves or rejects the request.

6.3 Rejection Path

If the approver rejects the request, the workflow updates the request to the appropriate rejected state and sends a rejection notification to the requester.

6.4 Fulfillment Path

If approved, the workflow creates a fulfillment task for the responsible team. The team processes the rental requirement and completes the task. The workflow then moves the request toward completion.

6.5 Completion

When fulfillment is complete, ServiceNow updates the request status and sends a completion notification to the requester.

7. Roles and Responsibilities

- Requester – Enters accurate rental information and submits the request.
- Approver – Reviews the request and approves or rejects it.
- Fulfillment/Rental Team – Processes approved requests and completes assigned tasks.
- ServiceNow Administrator – Configures the catalog item, workflow, notifications, access, and maintenance.

8. Notifications

- Request submitted – Confirms that the request was received.
- Approval required – Alerts the approver that action is required.
- Request approved – Informs the requester that the request has been approved.
- Request rejected – Informs the requester that the request was rejected.
- Fulfillment/completion – Informs the requester when processing is completed.

9. Non-Functional Requirements

- Usability – The catalog form should be simple and easy to understand.
- Availability – The solution should be available whenever the ServiceNow instance is available.
- Security – Only authorized users should access request and approval information.
- Auditability – Request, approval, and task activities should be traceable through ServiceNow records and history.
- Maintainability – Workflow and catalog configuration should be structured so that future changes can be managed easily.

10. Error and Exception Handling

- Missing mandatory information should prevent request submission.
- Invalid date combinations should be rejected according to configured validation rules.
- Rejected approvals should follow the defined rejection path.
- Workflow or task exceptions should be monitored by the ServiceNow administrator.
- Users should receive appropriate status information when processing cannot continue normally.

11. Testing Requirements

- Verify successful catalog request submission.
- Verify mandatory field validation.

- Verify workflow initiation.
- Verify approval notification and approval decision.
- Verify rejection path and requester notification.
- Verify fulfillment task creation and assignment.
- Verify task completion and request closure.
- Verify requester notifications at each configured stage.
- Verify request status and history tracking.

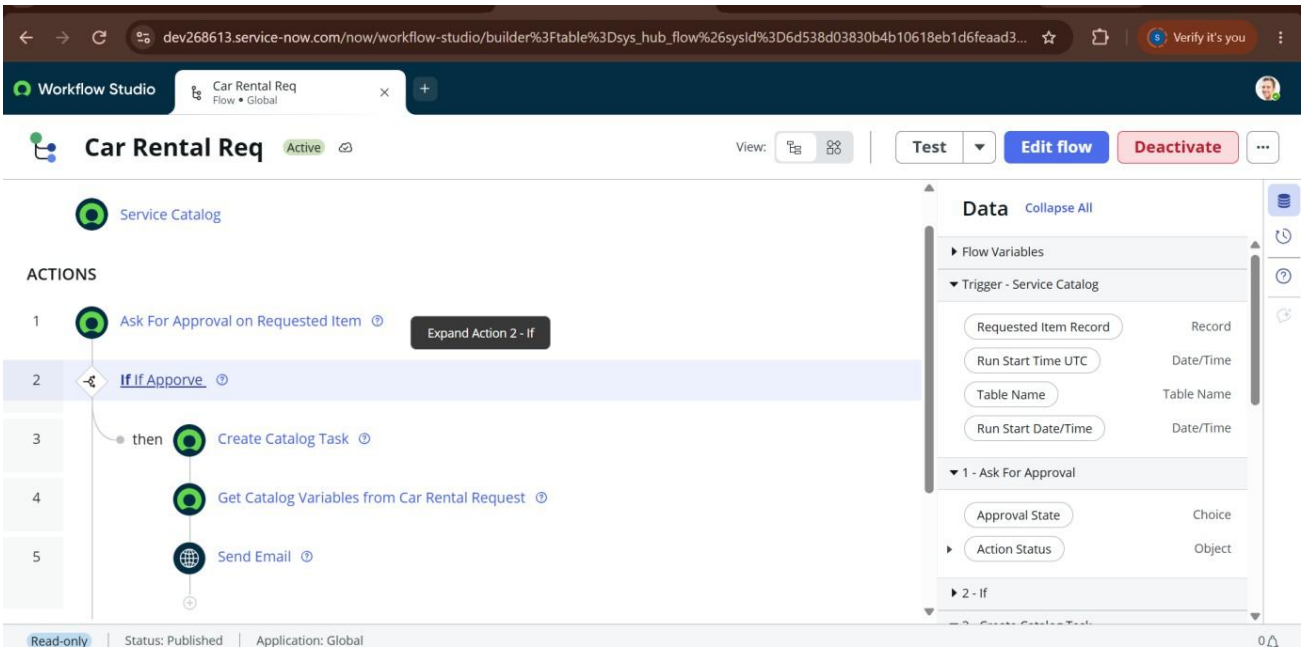
12. Assumptions and Dependencies

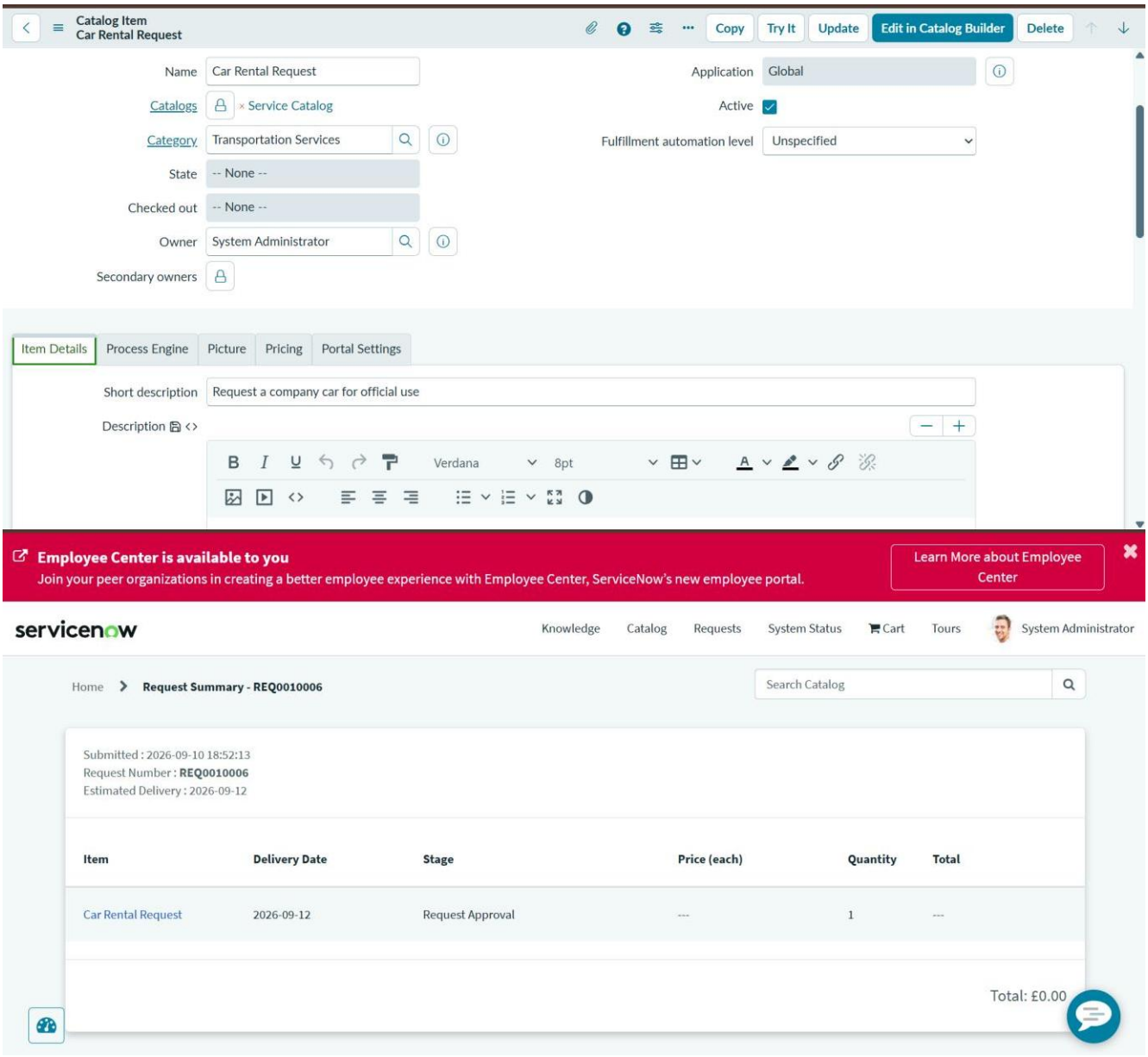
- The ServiceNow instance and required Service Catalog/Workflow functionality are available.
- Approvers and fulfillment groups are configured in ServiceNow.
- Business rules for rental dates, vehicle types, and approvals are defined by the organization.
- Notification/email functionality is available in the ServiceNow environment.

13. Future Enhancements

- Integrate with an external fleet or car rental system.
- Automatically check vehicle availability.
- Add rental cost estimation and budget approval.
- Add automated reminders for pending approvals.
- Add dashboards for request volume, approval time, and fulfillment performance.
- Add SLA-based monitoring for delayed fulfillment.

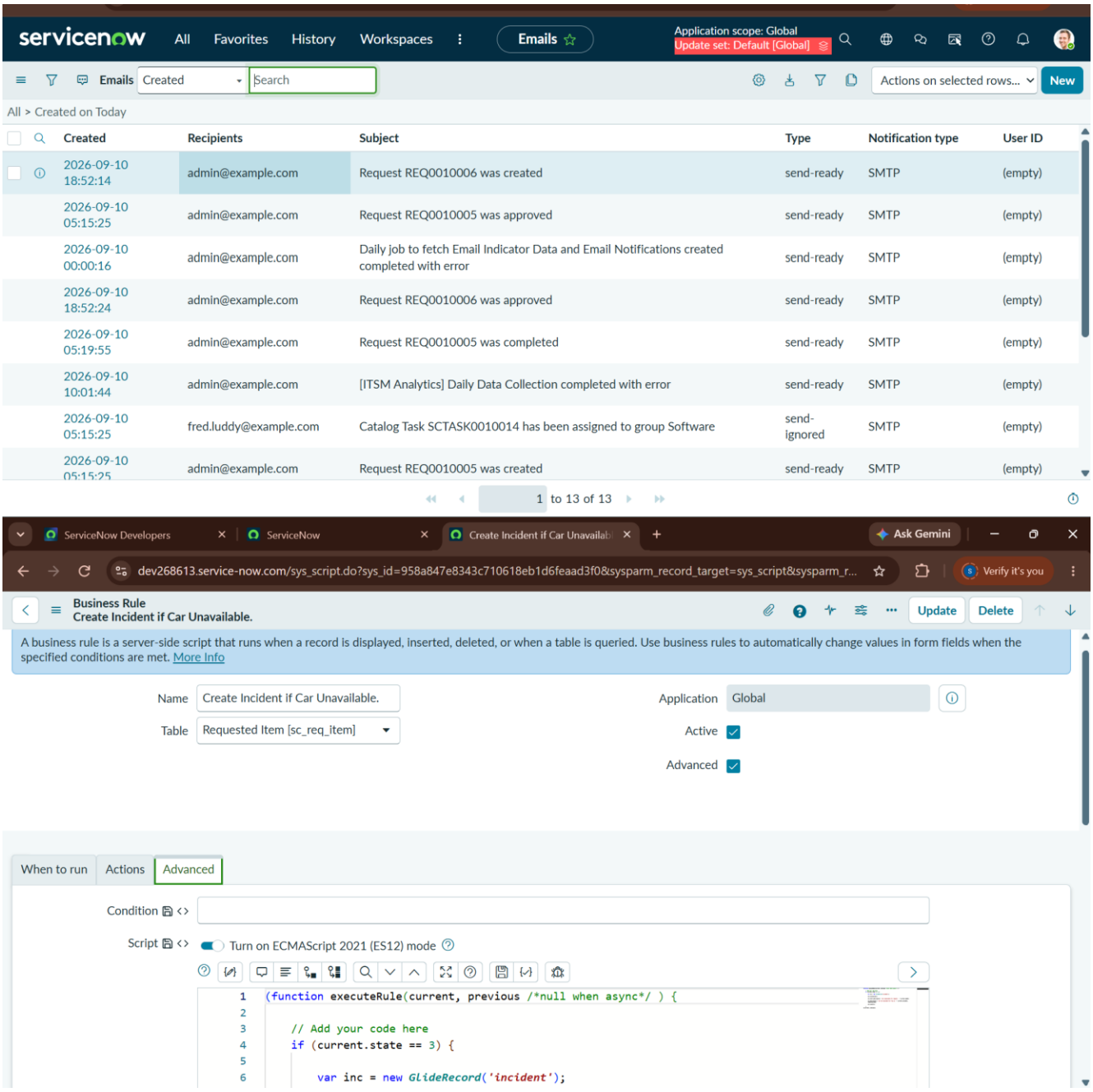
14. Results




Catalog Item
Car Rental Request
Copy
Try It
Update
Edit in Catalog Builder
Delete
Name
Car Rental Request
Application
Global
Catalogs
Service Catalog
Active
Category
Transportation Services
Fulfillment automation level
Unspecified
State
-- None --
Checked out
-- None --
Owner
System Administrator
Secondary owners
Item Details
Process Engine
Picture
Pricing
Portal Settings
Short description
Request a company car for official use
Description
Verdana
8pt
Employee Center is available to you
Join your peer organizations in creating a better employee experience with Employee Center, ServiceNow's new employee portal.
Learn More about Employee Center
servicenow
Knowledge
Catalog
Requests
System Status
Cart
Tours
System Administrator
Home
Request Summary - REQ0010006
Search Catalog
Submitted : 2026-09-10 18:52:13
Request Number : REQ0010006
Estimated Delivery : 2026-09-12
Item
Delivery Date
Stage
Price (each)
Quantity
Total
Car Rental Request
2026-09-12
Request Approval

1

Total: £0.00



15. Appendix

https://github.com/NVkishore22/Car-Rental-Request-Automation-ServiceNow